

Circle Equations in Context

Writing, reading and completing the square on circle equations from real site plans

- Every map in this set has its axes in the units named by the problem, and every centre-radius form is $(x - h)^2 + (y - k)^2 = r^2$.
 - Leave answers exact unless a problem asks you to round.
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1. A circular splash pad is being marked out on a park map whose axes are in metres. Its centre is at $(4, 7)$, and a drinking fountain at $(4, 14)$ sits exactly on its edge. Write the equation of the splash pad's boundary in centre-radius form.

Answer: _____

2. A cell tower's coverage area is modelled by $(x + 5)^2 + (y - 2)^2 = 81$, where x and y are kilometres east and north of the town hall.

(a) Find the radius of the coverage area.

Answer: _____ km

(b) A farm stands at $(4, 2)$. Evaluate the left-hand side of the equation at the farm's position.

3. A circular garden bed is planned so that the two endpoints of one diameter fall at $(1, 3)$ and $(9, 9)$ on the site plan, with axes in metres.

(a) Find the centre of the bed. _____

(b) Find the radius of the bed.

Answer: _____ m

(c) Write the equation of the bed's boundary in centre-radius form. _____

4. A drone operator's circular no-fly zone satisfies $x^2 + y^2 - 6x + 10y + 18 = 0$, with x and y measured in kilometres from the control tower.

(a) Rewrite the equation in centre-radius form. _____

(b) Find the radius of the no-fly zone.

Answer: _____ km

5. The edge of a lighthouse's visible range is the circle $(x - 12)^2 + (y + 5)^2 = 169$, with x and y in kilometres east and north of the harbour office. A ship is at $(0, 7)$.

(a) Evaluate the left-hand side of the equation at the ship's position. _____

(b) The lighthouse stands at the centre. Find the distance from the lighthouse to the ship, to the nearest hundredth.

Answer: _____ km

(c) A point is on the range boundary exactly when the left-hand side equals 169. Write $<$, $=$ or $>$ to compare your answer to part (a) with 169, then state whether the ship is inside, on, or outside the range. _____

6. A circular running track on a park map (axes in metres) has its centre at $(6, 2)$ and is tangent to a straight fence that runs along the line $y = -4$.

(a) Find the radius of the track.

Answer: _____ m

(b) Write the equation of the track in centre-radius form. _____

7. A city's flood-warning zone is the circle $x^2 + y^2 + 14x - 8y - 16 = 0$, with x and y in kilometres from the city centre, which sits at the origin.

(a) Rewrite the equation in centre-radius form. _____

(b) Find the radius of the zone.

Answer: _____ km

(c) Find the distance from the origin to the centre of the zone, to the nearest hundredth, then explain whether the city centre lies inside the zone. _____

8. Two water mains meet the boundary of a circular reservoir at $A(2, 9)$ and $B(10, 3)$, and $\overline{AB}$ is a diameter. The axes are in metres. The reservoir is to be rebuilt with the same centre but a radius one and a half times as large.

(a) Find the centre of the reservoir. _____

(b) Find the new radius.

Answer: _____ m

(c) Write the equation of the rebuilt boundary in centre-radius form. _____